

CSMFAB000000000040 Smart Rope Actuator Design Specification

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | **Date:** June 2026 **Category:** Space Launch / Production

Revision Drivers (all V2.0 documents)

- 1. **2025-2026 experimental and regulatory data** supersedes V1.0 specifications
- 2. **Solar Cycle 25 SSN ~137 peak** (observed) — elevated urgency for all Carrington-focused work
- 3. **Standards updates** — AASHTO 2024, IMO 2025, IEEE P2945 draft, DNV GL EMP-2, ClassNK FRP

V1.0 vs V2.0 Parameter Changes

Parameter	V1.0	V2.0	Consequence if Not Updated
Connector type	Threaded PEEK (5 turns)	Bayonet-lock ZrO2 (quarter turn)	8,000 thread engagements = 4-5hr assembly; bayonet = 7 min
Segment cost	\$18/segment (commercial ZrB2)	\$9.50/segment (in-house SHS ZrB2)	47% cost reduction via Aegis-Forge I supply chain
Deployment speed	1.2 ms per segment	0.8 ms (MAX Phase hinge)	33% faster full-rope deployment
Operating temp range	-50 to +200degC	-60 to +400degC	Extended for re-entry thermal scenario
Quality control	Manual inspection	LIBS auto-analysis (Aegis-Forge I)	Manual misses sub-surface ZrB2 phase defects

Unchanged Elements

All core design philosophy, threat physics (GIC induction equations), material selection rationale, and fundamental architecture are carried forward unchanged from V1.0. Only the parameters listed above were revised.

End Versioning Delta | CSMFAB000000000040 Smart Rope Actuator Design Specification | June 2026